Navin Sridhar

Contact info
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Employment

Stanford University, CA
- Simons Collaboration (SCEECS) postdoctoral fellowship/KIPAC (2024–2028)
- Supervisor: Dr. Roger Blandford

Education

Columbia University, NY
- Ph.D. in Astrophysics (May 2024)
- M.A. (2020), M.Phil (2021) in Astrophysics
  Advisors: Dr. Lorenzo Sironi, Dr. Brian D. Metzger
  Thesis: Electromagnetic and Multimessenger Signals From Magnetized Outflows of Compact Object Binaries

Indian Institute of Science Education & Research, Bhopal
- BS-MS in Physics (CPI: 9.05/10) (May 2018)
  Thesis: Accretion disk evolution around black holes, and thermonuclear bursts in neutron stars. (Caltech and TIFR)

Awards and fellowships

1. PI of NASA NICER grant (US$45,000) (2026–2027)
2. Neil Gehrels Prize Fellowship: JSI, University of Maryland & NASA (2024)
3. Brinson Fellowship: Northwestern University/CIERA & University of Chicago (2024)
4. NASA *FINESST* award (US$100,000) (2022–2024)
5. PI of NASA NICER grant (US$37,500) (2022–2024)
6. Simons Foundation Aspen center for physics grant (US$1,350) (July 2023)
7. International Astronomical Union (S378) travel grant (US$1,600) (March 2023)
8. Columbia Arts and Sciences Graduate Council travel grant (US$500) (2022)
9. Columbia ASGC Student/Diversity Initiative Grant (US$2,000) (2022–2023)
10. AAS National Osterbrock Leadership fellow (US$6,400) (2021–2023)
11. The Yeh Family endowment fellowship, Columbia University (2019–2020)
12. Dean's fellowship for graduate study at Columbia University (2018–2020)
13. Caltech Summer Undergraduate Research Fellowship (US$6,600) (July–Oct 2017)
14. Mitacs Globalink Research Internship Award (∼CA$6,600) (May–July 2017)
15. *INSPIRE* fellowship by DST, Govt. of India (INR 400,000) (2013–2018)

Research Experience/Visits

- Affiliated scientist, SCEECS (2023–2027)
  Simons Collaboration on Extreme Electrodynamics of Compact Sources
- Visiting Researcher, **KITP**, UC Santa Barbara (Sept 2024)
- Visiting Student Researcher, **Caltech**, Pasadena (2023–2024)
  Host: Dr. Sterl Phinney
- Stellar Interactions and the Transients they Cause, **Aspen** (July 2023)
  Host: Aspen Center for Physics, Colorado
- Visiting researcher, Simons Foundation – plasma astrophysics program (2019–2023)
  Center for Computational Astrophysics, **Flatiron Institute**, NY
  Advisor(s): Dr. Sasha Philippov, Dr. Lorenzo Sironi
- Summer Undergraduate Research Fellow (SURF), **Caltech**, Pasadena (July–Oct 2017)
  Advisor(s): Dr. Javier García, Prof. Fiona Harrison
- Mitacs Globalink Research Intern, **University of Alberta**, Edmonton (May–July 2017)
  Advisor: Dr. Rodrigo Fernández
- Visiting Summer Research Program, **TIFR**, Mumbai (May–July 2016)
  Advisor: Prof. Sudip Bhattacharyya
- Indian Academy of Sciences summer research fellow, **ISRO**, Bangalore (June–Aug 2015)
  Advisor: Dr. Anuj Nandi
- Visiting Student Program, **IIAp**, Bangalore (May–Dec 2015)
  Advisor: Prof. Aruna Goswami

Publications (Total: 41) NASA ADS | Google Scholar

As a lead author (10):

1. **Sridhar, N.**, Sobacchi, E., Sironi, L., et al. 2026. "*Interaction of strong electromagnetic waves with unmagnetized pair plasmas*" Physical Review Letters 137, 045102.
2. **Sridhar, N.**, Ripperda, B., Sironi, L., et al. 2025. "*Bulk Motions in the Black Hole Jet Sheath as a Candidate for the Comptonizing Corona*" ApJ 979, 199
3. **Sridhar, N.**, Metzger, B. D., & Fang, K. 2024. "*High-Energy Neutrinos from Gamma-Ray-Faint Accretion-Powered Hypernebulae.*" ApJ 960, 74
4. **Sridhar, N.**, Sironi, L., & Beloborodov, A. M. 2023. "*Comptonization by Reconnection Plasmoids in Black Hole Coronae II: Electron-Ion Plasma.*" MNRAS 518, 1301-1315
5. **Sridhar, N.**, & Metzger, B. D. 2022. "*Radio Nebulæ from Hyper-Accreting X-ray Binaries as Common Envelope Precursors and Persistent Counterparts of FRBs*" ApJ 937, 5
6. **Sridhar, N.**, Sironi, L., & Beloborodov, A. M. 2021. "*Comptonization by Reconnection Plasmoids in Black Hole Coronae I: Magnetically Dominated Pair Plasma*" MNRAS 507, 5625-5640
7. **Sridhar, N.**, Metzger, B. D., Beniamini, Paz., et al. 2021. "*Periodic Fast Radio Bursts from Luminous X-ray Binaries.*" ApJ 917, 13
8. **Sridhar, N.**, Zrake, J., Metzger, B. D., et al. 2021. "*Shock-powered radio precursors of neutron star mergers from accelerating relativistic binary winds.*" MNRAS 501, 3184-3202
9. **Sridhar, N.**, García, J. A., Steiner, J. F., et al. 2020. "*Evolution of the Accretion Disk—Corona during the Bright Hard-to-soft State Transition: A Reflection Spectroscopic Study with GX 339–4.*" ApJ 890, 53
10. **Sridhar, N.**, Bhattacharyya, S., Chandra, S., & Antia, H. M. 2019. "*Broad-band reflection spectroscopy of MAXI J1535–571 using AstroSat: estimation of black hole mass and spin.*" MNRAS 487, 4221-4229

$\leq 3^{\text{rd}}$ author, with significant contributions (10): [† = supervised student]

11. Uno, K.†, Metzger, B. D., **Sridhar, N.** et al. 2025. "*The Population of Radio Hypernebulae from Mass-Transferring Massive Binaries*" to be submitted to ApJ
12. Gupta, S.†, **Sridhar, N.**, Sironi, L. 2023. "*Comptonization by Reconnection Plasmoids in Black Hole Coronae III: Dependence on the Guide Field in Pair Plasma.*" MNRAS 527, 6065–6075
13. Gupta, S.†, **Sridhar, N.**, Sironi, L. 2025. "*The Role of Electric Dominance for Particle Injection in Relativistic Reconnection*" MNRAS 538, 49-59
14. Bhandari, S., Marcote, B., **Sridhar, N.**, et al. 2023. "*Constraints on the persistent radio source associated with FRB 20190520B using the European VLBI Network*", ApJ 958, 2
15. Eftekhari, T., Fong, W-F., Gordon, A. C., **Sridhar, N.**, et al. 2023. "*An X-ray census of Fast Radio Burst host galaxies: constraints on AGN and X-ray counterparts*", ApJ 958,1
16. Metzger, B. D., **Sridhar, N.**, Margalit, B., et al. 2022. "*A Toy Model for the Time-Frequency Structure of Fast Radio Bursts: Implications for the CHIME/FRB Burst Dichotomy.*" ApJ 925, 135
17. Margalit, B., Beniamini, P., **Sridhar, N.**, Metzger, B. D., 2020. "*Implications of a Fast Radio Burst from a Galactic Magnetar.*" ApJL 899, L27
18. García, J. A., Tomsick, J. A., **Sridhar, N.**, et al. 2019. "*The 2017 Failed Outburst of GX 339–4: Relativistic X-Ray Reflection near the Black Hole Revealed by NuSTAR and Swift Spectroscopy.*" ApJ 885, 48
19. Bhattacharyya, S., Yadav, J. S., **Sridhar, N.**, et al. 2018. "*Effects of Thermonuclear X-Ray Bursts on Non-burst Emissions in the Soft State of 4U 1728-34.*" ApJ 860, 88
20. Karinkuzhi, D., Goswami, A., **Sridhar, N.**, et al. 2018. "*Chemical analysis of three barium stars: HD 51959, HD 88035, and HD 121447.*" MNRAS 476, 3086-3096

$> 3^{\text{rd}}$ author (16):

21. Shuvo, P., Meyer, E., ⋯, **Sridhar, N.**, et al. 2026. "*First Detection of Radio Polarization During Jet Formation in the Changing-Look AGN 1ES 1927+654*" arXiv:2700.00000; Submitted to ApJ.
22. Uttarkar, P., Shannon, R., ⋯, **Sridhar, N.**, et al. 2026. "*A fast radio burst cyclone in technicolour: evidence of plasma lensing*" arXiv:2602.16409; Submitted to Nature.

23. Moroianu, A., Bhandari, S., ⋯, **Sridhar, N.**, et al. 2026. "*A Milliarcsecond Localization Associates FRB 20190417A with a Compact Persistent Radio Source and an Extreme Magnetoionic Environment*" ApJL 996, L16
24. Snelders, M. P., Hessels, J. W. T., ⋯, **Sridhar, N.**, et al. 2025. "*Revisiting FRB 20121102A: milliarcsecond localisation and a decreasing dispersion measure*" arXiv:2510.11352
25. Blanchard, P., Berger, E., ⋯, **Sridhar, N.**, et al. 2025. "*James Webb Space Telescope Observations of the Nearby and Precisely Localized FRB20250316A: A Potential Near-IR Counterpart and Implications for the Progenitors of Fast Radio Bursts*" ApJL 989, L49
26. Ibik, A., Drout, M., ⋯, **Sridhar, N.**, et al. 2024. "*A search for persistent radio sources toward repeating fast radio bursts discovered by CHIME/FRB*" ApJ 976, 199
27. Dong, Y., Eftekhari, T., ⋯, **Sridhar, N.**, et al. 2024. "*A Radio Study of Persistent Radio Sources in Nearby Dwarf Galaxies: Implications for Fast Radio Bursts*", ApJ 973, 133
28. Sarin, N., Clarke, T. A., ⋯, **Sridhar, N.** 2024. "*The origin of the coherent radio flash potentially associated with GRB 201006A*", ApJL 973, L20
29. AXIS TDAMM SWG., ⋯, incl. **Sridhar, N.**, et al. 2023. "*Prospects for Time-Domain and Multi-Messenger Science with AXIS*", Universe, 10, 316
30. Connors, R. M. T., Tomsick, J. T., ⋯, **Sridhar, N.**, et al. 2023. "*The High Energy X-ray Probe (HEX-P): Probing Accretion onto Stellar Mass Black Holes*", Front. Astron. Space Sci. 10:1292682
31. Kammoun, E., Lohfink, A. M., ⋯, **Sridhar, N.**, et al. 2023. "*The High Energy X-ray Probe (HEX-P): Probing the physics of the X-ray corona in active galactic nuclei*", Front. Astron. Space Sci. 10:1308056
32. Dong, Y., Eftekhari, T., ⋯, **Sridhar, N.**, et al. 2023. "*Mapping Obscured Star Formation in the Host Galaxy of FRB 20201124A*", ApJ 961, 44
33. Bhandari, S., Gordon, A. C., ⋯, **Sridhar, N.**, et al. 2023. "*A non-repeating fast radio burst in a dwarf host galaxy*", ApJ 948, 67
34. Connors, R. M. T., García, J. A., ⋯, **Sridhar, N.**, et al. 2021. "*Reflection Modeling of Black Hole Binary 4U 1630–47: Disk Density and Returning Radiation*", ApJ 909, 146
35. Connors, R. M. T., García, J. A., ⋯, **Sridhar, N.**, et al. 2020. "*Evidence for Returning Disk Radiation in the Black Hole X-Ray Binary XTE J1550–564.*" ApJ 892, 47
36. Connors, R. M. T., García, J. A., ⋯, **Sridhar, N.**, et al. 2019. "*Conflicting Disk Inclination Estimates for the Black Hole X-Ray Binary XTE J1550–564.*" ApJ 882, 179
37. Verdhan Chauhan, J., Yadav, J. S., ⋯, **Sridhar, N.**, et al. 2017. "*AstroSat/LAXPC Detection of Millisecond Phenomena in 4U 1728-34.*" ApJ 841, 41

Books and white papers (2):

38. Koss, M., ·, incl. **Sridhar, N.**, et al. 2025. "*The Advanced X-ray Imaging Satellite Community Science Book*", arXiv:2511.00253
39. Safi-Harb, S., Burdge, K., ⋯, incl. **Sridhar, N.**, et al. 2023. "*From Stellar Death to Cosmic Revelations: Zooming in on Compact Objects, Relativistic Outflows and Supernova Remnants with AXIS*", arXiv:2311.07673

Astronomical Telegrams (2):

40. Blanchard, P., Berger, E., ⋯, **Sridhar, N.**, et al. 2015. "*Deep JWST/NIRCam Imaging of FRB 20250316A: Detection of Potential IR Counterparts.*" ATel #17195
41. García, J. A., Harrison, F., ⋯, **Sridhar, N.**, et al. 2017. "*NuSTAR Observation of GX 339–4 in the early stages of its 2017 outburst.*" ATel #10825

OBSERVING EXPERIENCE

X-ray: 15 Ms; Radio: 142 ks; Optical: 65 ks; IR: 4.5 hr
Total grant: $624,935
As Principal Investigator:

1. NICER archival data of transitional ms pulsar (Cycle AO8; 1.2 Ms of data; US$45,000).
2. Joint NICER and NuSTAR observations studying reflection features from X-ray binaries (Cycle AO3; 60 ks; US$37,500).
3. uGMRT multi-band radio observations of Giant Radio Pulses (Cycle 34; 36 ks).
4. *AstroSat* observations of ∼10 sources: accretion powered millisecond pulsars, neutron star type-I bursts, black hole X-ray binaries (Cycles A04-A08; cumulatively >300 ks).

As Co-Investigator:

1. JWST search for a transient NIR counterpart to FRB 20250316A (PI: Peter Blanchard; $110,500 ; cycle 5, 2026).
2. JWST observation of FRB 20250316A (PI: Peter Blanchard; cycle 4, 2025).
3. Joint NuSTAR and NICER observation to study the reflection properties and evolution of accretion disks in black hole binaries (Cycle 10; 500 ks; PI: J. Garcia).
4. GBT observations to probe the dynamic environment of FRB 20180301A (Cycle 2024B; 108 ks; PI: P. Kumar).
5. Chandra observations to probe the engines of fast radio bursts (Cycle 25; 106 ks; PI: T. Eftekhari).
6. GBT observations to probe the local environment of FRB 20180301A (Cycle 2023B; 72 ks; PI: P. Kumar).
7. VLA observations to constrain FRB 'persistent radio sources' (Cycle 2023B; 10 ks; PI: Y. Dong).
8. NICER observations for amassing a comprehensive spectral timing archive of outbursting black holes (Cycle AO5; 200 ks; $43,000; PI: J. Steiner).
9. Joint NICER and NuSTAR observation to study the reflection properties and evolution of accretion disks in black hole binaries (Cycle AO4; 110 ks; $44,000 ; PI: J. Garcia).
10. Joint NICER and NuSTAR observation of XRBs during its outburst (Cycle AO4; 88 ks; $44,000; PI: R. Connors).
11. NICER observations for black hole spin measurements using continuum fitting (Cycle AO4; 40 ks; $37,500; PI: J. Steiner).
12. Joint NICER and NuSTAR observations for measuring black hole spin and mass through X-ray reflection spectra and reverberation lags (Cycle AO4; 30 ks; PI: G. Mastroserio).
13. Joint XMM-Newton and NuSTAR observations for studying truncated accretion disks in black holes (AO 21; 120 ks; $50,656; PI. J. García).
14. Chandra observation of nearby Fast Radio Bursts (Cycle 23; 55 ks; $52,519; PI: T. Eftekhari).
15. Joint NICER and NuSTAR observation of XRBs during its outburst (Cycle AO3; 88 ks; $37,500; PI: R. Connors).
16. NICER observations for black hole spin measurements using continuum fitting (Cycle AO3; 40 ks; $44,000; PI: J. Steiner).
17. Joint NICER and NuSTAR observations for measuring black hole spin and mass through X-ray reflection spectra and reverberation lags (Cycle AO3; 30 ks; $37,500; PI: G. Mastroserio).
18. NICER observations monitoring the evolving thermal disk emission of LMC X-3 (Cycle AO2; 60 ks; $20,229; PI: J. Steiner).
19. Joint NICER and NuSTAR observation of XRBs during its outburst (Cycle AO2; 90 ks; $21,031; PI: R. Connors).

Others:

1. Mt. Palomar 200" Hale telescope: DBSP observations of 14 variable AGN (2 nights)
2. NRAO Green Bank Telescope: radio pulsar observations (24 ks)

Delivered talks/presentations (*29 invited)

83. Talk and session chair at the FRB 2026 conference, **Guiyang**, China (July 2026)
82. *Seminar at Department of Astronomy, **Tsinghua University**, Beijing (July 2026)
81. Talk at the BlackHolistic 2026 conference, **Oxford University**, UK (March 2026)
80. *Astrophysics seminar, **Washington University**, St. Louis, MO (Oct 2025)
79. 22nd AAS-High Energy Astrophysics Division meeting, St. Louis, MO (Oct 2025)
78. *Black-board talk, Frontiers of Relativistic Plasma Physics, **KITP**, USA (Sep 2025)
77. *Overview of FRB emission mechanisms, FRB 2025 conference, **McGill**, Canada (July 2025)
76. Seminar, **Inter-University Center for Astronomy & Astrophysics**, Pune (June 2025)
75. *Astrophysics colloquium, **Tata Institute of Fundamental Research**, Mumbai (May 2025)
74. Astrophysics and Relativity seminar, **International Center for Theoretical Sciences**, Bangalore (May 2025)
73. Relativistic Magnetospheres workshop, **Ecole De Physique Des Houches**, France (April 2025)
72. Magnetic fields around Compact Objects workshop, **Perimeter Institute**, Canada (March 2025)
71. Fast Radio Burst Frontiers workshop, **Carnegie Mellon University**, USA (March 2025)
70. *Talk at the XOC group meeting **Stanford**, CA (Sept 2024)

69. Talk at the High-energy plasma phenomena in astrophysics workshop **MIAPbP, Munich**, Germany (Sept 2024)
68. Graduating student colloquium, **Columbia University**, NY (April 2024)
67. **_AXIS_* seminar series (April 2024)
66. 21st AAS-High Energy Astrophysics Division meeting, Texas (April 2024)
65. *Seminar, Simons Collaboration on Extreme Electrodynamics of Compact Object Sources (**SCEECS**) (Feb 2024)
64. *Astroplasmas seminar, **Princeton University**, NJ (Dec 2023)
63. *CIERA Theory seminar, **Northwestern University**, IL (Nov 2023)
62. JSI workshop on 'Winds throughout the Universe', **Annapolis**, MD (Oct 2023)
61. *Theoretical astrophysics group board talk, **Carnegie Observatories**, CA (Sept 2023)
60. *Seminar, **Syracuse University**, NY (Sept 2023)
59. *Review talk at Astrophysics of Fast Radio Bursts II workshop (+session chair), **CCA, Flatiron Institute**, NY (Sept 2023)
58. *Colloquium, **National Center for Radio Astrophysics**, Pune, India (June 2023)
57. *Astrophysics seminar, **Tata Institute of Fundamental Research**, Mumbai(May 2023)
56. *'Transient Tuesday' talk, DARK, Niels Bohr Institute **Copenhagen**, Denmark(May 2023)
55. *Discussion chair, Fast Radio Burst Follow-Up workshop, **University of Toronto** (April 2023)
54. *High energy astrophysics seminar, **Hebrew University of Jerusalem**, Israel(March 2023)
53. *Seminar, Astrophysics Research Center of **The Open University**, Israel (March 2023)
52. *High energy astrophysics meeting, Black Hole Initiative, **Harvard**, MA (Feb 2023)
51. Cosmic transients lab seminar, Center for Astrophysics, **Harvard**, MA (Feb 2023)
50. High Energy Astrophysics seminar, Center for Astrophysics, **Harvard**, MA (Feb 2023)
49. FRB Taiwan 2023, National Chung Hsing University, Taichung **Taiwan** (Feb 2023)
48. *Workshop on 'Overcoming disconnects in the understanding of accreting black holes', **Lorentz Center, Leiden**, The Netherlands (Jan 2023)
47. HEX-P black hole binaries, **science working group** presentation (Dec 2022)
46. *CCAPP seminar, **Ohio State University**, OH (Nov 2022)
45. 64th APS Division of Plasma Physics annual meeting, **Spokane**, WA (Oct 2022)
44. Review talk on Observations of FRBs, **CCA, Flatiron Institute**, NY (Oct 2022)
43. **AXIS* 'compact objects and supernova remnants' **science working group** (Oct 2022)
42. *Astrophysics seminar, **Cornell University**, NY (Sept 2022)
41. *AGN seminar at **NASA** Goddard Space Flight Center, MD (Sept 2022)
40. **NASA** Time Domain and Multi-Messenger (TDAMM) Initiative Workshop, **Annapolis**, MD (Aug 2022)
39. FRB 2022 meeting, IAUGA, **Busan**, South Korea (Aug 2022)
38. Summer BLAST workshop, Black Hole Initiative, **Harvard**, MA (July 2021)
37. *AXIS* 'Time-Domain and Multi-Messenger Astronomy' **science working group** (July 2022)
36. Special pizza lunch seminar, **Caltech**, CA (June 2022)
35. Seminar at **Carnegie Observatories**, CA (June 2022)
34. 240th meeting of the American Astronomical Society, **Pasadena**, CA (June 2022)
33. Workshop on Relativistic Plasma Astrophysics, **Purdue University**, IN (May 2022)
32. Compact objects group meeting at **CCA, Flatiron Institute**, NY (April 2022)
31. *Kapteyn Astronomical Institute, **Groningen**, The Netherlands (April 2022)
30. 19th AAS-High Energy Astrophysics Division meeting, **Pittsburgh**, PA (March 2022)
29. Brown bag lunch seminar, **Massachusetts Institute of Technology**, MA (Feb 2022)
28. Winter BLAST workshop, Black Hole Initiative, **Harvard**, MA (Dec 2021)
27. Gothamfest, **CCA, Flatiron Institute**, NY (Dec 2021)
26. Kavli-IPMU workshop on particle acceleration by reconnection, **Tokyo**, Japan (Nov 2021)
25. 9th Microquasar workshop, **Cagliari**, Italy (Sept 2021)
24. Astrofest, **Columbia** University, NY (Sept 2021)
23. High-energy plasma phenomena in astrophysics, **Munich**, Germany (July 2021)
22. FRB 2021 conference (online) (July-Aug 2021)
21. *Princeton astro-coffee, **Princeton**, NJ (July 2021)
20. 'Midwest Magnetic Fields' Workshop, **Wisconsin-Madison**, (June 2021)
19. *CHIME/FRB collaboration, **McGill**, Canada (May 2021)
18. *Seminar at the **Institute of Astrophysics - FORTH**, Greece (April 2021)
17. Compact objects group meeting at **CCA, Flatiron Institute**, NY (March 2021)
16. **Kyoto** FRB meeting, **Yukawa Institute for Theoretical Physics** (Feb 2021)

15. Compact object group meeting, **TIFR**, Mumbai (Dec 2020)
14. 62nd Annual Meeting of the **APS Division of Plasma Physics** (Nov 2020)
13. Compact objects group meeting, **CCA, Flatiron Institute**, NY (June, Oct, Nov 2020)
12. Plasma astrophysics summer program, **CCA, Flatiron Institute**, NY (July 2019)
11. Theoretical High Energy Astrophysics seminar, **Columbia University** (June 2019)
10. Graduate Seminar, **Columbia University**, NY (Feb, April 2019)
9. Research Seminar, **Columbia University**, NY (Dec 2018)
8. Pizza lunch talk, **Columbia University**, NY (Oct 2018,2022)
7. *NuSTAR* SOC group meeting, **Caltech**, CA (Sept 2017)
6. Group presentation at **NRAO Green Bank Observatory**, WV (Sept 2017)
5. 16th AAS-High Energy Astrophysics Division meeting, **Sun Valley**, ID (March 2022)
4. Visiting Student Program ceremony, **University of Alberta**, Canada (July 2017)
3. Lunch talk at **University of Alberta**, Canada (June 2017)
2. 'Wide band X-ray spectro-temporal studies', **TIFR**, Mumbai, India (Jan 2017)
1. **IISER Bhopal** Physics department symposium (Nov 2015)

Mentoring

Total: 9 students

– *Ningyue Fan*; **Ph.D.** student, Stanford University (Fall 2025-)
– *Christopher Farias*; **undergraduate** student, San Diego State University (Summer 2025)
– *Kohki Uno*; **Ph.D.** student, Kyoto University (1 paper) → postdoc at Columbia University (Summer 2024)
– *Arnab Kundu*; **masters** thesis student, IISER Kolkata (2021–2022)
– *Akash Vani*; Pune University → Masters at Heidelberg University → Ph.D. student at MPA, Garching (2019–2020)

At Columbia University:

– *Sanya Gupta*; **undergraduate** research projects (2 papers) → Ph.D. student at Stanford University (2022–2025)
– *Shifra Mandel*; **graduate** mentor-mentee program (2021–2022)
– *Shuhan Zhang*; Women in Science at Columbia **undergraduate** mentorship program → Ph.D. student at UC San Diego (2021)
– *Bhairavi Chandrasekhar*; **undergraduate** summer internship (2020)

Teaching

At Columbia University:

– **Head TA** of the Astronomy department (2020–2021)
– **Instructor** for undergraduate Intro to Astronomy lab courses (Fall 2019, Spring 2020)
– **Teaching Assistant** for 2 undergraduate (non-major) courses (Fall 2018, Spring 2019)
– **Guest lectured** for ASTR UN1403 (in lieu of Prof. David Helfand) (Feb 2020)
– **Advanced track** in Columbia CTL's Teaching Development Program (TDP) (2018–2024)
– Completed CTL's Innovative Course Design Seminar (2021)

Outside Columbia University:

– Selected as a **Lead instructor** at the Yale Young Global Scholars (YYGS) program
Yale University, New Haven, CT (Summer 2020; postponed due to Covid-19)
– Trained to incorporate **inquiry-based learning techniques** in STEM education through the ISEE—Professional Development Program (PDP), **UC Santa Cruz** (March–Sept 2019)
– Designed and taught an inquiry activity on statistics, **UC Los Angeles** (Sept 2019)

Professional leadership

- Scientific Organizing Committee of symposium on *The birth, life and death of black holes*, European Astronomical Society annual meeting, Leiden, The Netherlands (June 2021)

At Stanford University:

- Postbaccalaureate program admissions committee member (2025)
- Ph.D. program admissions committee member (2024-2025)
- Started the weekly KIPAC Compact Objects Group meetings (Oct 2024–)

At Columbia University:

- Organized an eight-part lecture series titled *'Science Policy for Scientists'* (Spring 2023)
- Department representative, Arts and Sciences Graduate Council (2022-2023)
- Graduate representative, faculty hiring committee, Astronomy department (2021–2022)
- Graduate representative, colloquium committee, Astronomy department (2022-2023)
- Graduate admissions committee member, Astronomy department (2021–2022)
- Co-convener of *Astrofest* (Sept 2020)

At IISER Bhopal:

- Senate representative, Central Advisory Committee, Students Activity Council (2016-2017)
- Elected Science Council secretary (2014-2016)
- Convenor of *Singularity*, the Institute annual science festival (April 2015)
- Founder of the Institute *Astronomy Club* (2014)
- Founder of the Institute *Quiz Club* (2014)

PROFESSIONAL SERVICE

- Invited to serve as a **referee** for:
  –Journal of Plasma Physics,
  –Proceedings of the National Academy of Sciences of the USA,
  –Nature,
  –The Astrophysical Journal,
  –Monthly Notices of Royal Astronomical Society,
  –Physical Review X,
  –Astronomy & Astrophysics.
- Invited to serve as a **panelist** for:
  –NASA's Chandra, NICER observatories
  –National Science Foundation,
  –*AstroSat*,
  –NCRA-uGMRT,
  –Observer at Hubble Space Telescope time allocation committee.
- **Science working group member** of:
  –NewAthena (2025–)
  –Advanced X-ray Imaging Satellite (**AXIS**) (2022–)
  –High-Energy X-ray Probe (**HEX-P**) (2022–)
  –PRobe for far-Inrared Mission for Astrophysics (**PRIMA**) (2025–)
- DEI committee member, Columbia Arts and Sciences Graduate Council (2022-2023)
- **Judge**, Chambliss poster competition, American Astronomical Society meeting (2022)
- **Webmaster:** (1) Columbia Astronomy Department, (2) Theoretical High-Energy Astrophysics group, Columbia, (3) Simons Collaboration on Extreme Electrodynamics of Compact Sources
- Member of American Astronomical Society (2018–present)
- Member of New York Academy of Sciences (2018–2023)

SUMMER SCHOOLS

- **Multiscale modeling of astrophysical and space plasmas – 2019**
  Center for Computer Astrophysics (CCA), Flatiron Institute, New York, NY
- **Multiple approaches to plasma physics from laboratory to astrophysics – 2019**
  Ecole de Physique Des Houches, Les Houches, France
- **NRAO Green Bank Telescope observer training workshop – Fall 2017**
  Green Bank Observatory, West Virginia
- **Summer School on Gravitational Wave Astronomy – 2016**
  International Center for Theoretical Sciences (ICTS), TIFR, Bangalore
- **Summer school on relativity, black holes and neutron stars – 2015**
  Indian Institute of Astrophysics (IIAp), Bangalore

COMPUTING/ SOFTWARE EXPERIENCE

- Languages and tools: Python, git, `FORTRAN`, IRAF, gnuplot, DS9, LaTeX, *ftools* (HEASoft): `XSPEC`, `QDP`, `XRONOS`
- Supercomputers: NASA Pleiades, DoE-NERSC Cori, NSERC WestGrid, Flatiron Institute HPC cluster, Columbia University Habanero, Terremoto, Ginsburg clusters
- Numerical tools: Particle-In-Cell, Magnetohydrodynamical simulations
- Telescope data handling: *AstroSat*, *Swift*, *RXTE*/PCA, *NuSTAR*, *NICER*, *XMM*-Newton, NCRA-GMRT, NRAO-GBT (Astrid, CLEO)

SCIENCE OUTREACH

1. Public outreach talk during KIPAC community day, Stanford University (April 2025)
2. Research career in astrophysics for undergraduates at IISER Bhopal (Jan 2025)
3. KIPAC public outreach talk, Stanford University (Dec 2024)
4. Co-organized astronomy outreach activities at Columbia University for the South African academic achievers program (750 high-school students) (Oct 2022)
5. Talk on introduction/orientation to summer research, for undergraduate researchers at Columbia University (June 2022)

6. Popular talk on black hole X-ray binaries at IISER Bhopal Astronomy club (Oct 2021)
7. Mentored the students of Democracy Prep High School, Harlem, NYC, on the foundations of computer coding (Fall 2021)
8. Research career in astrophysics for undergraduates at IISER Bhopal (April 2021)
9. Delivered a lecture on science with space-based X-ray telescopes, for high-school students at *Frontiers of Science*—the annual outreach program at TIFR, Mumbai. (Nov 2017)
10. Organized a tour of GMRT for TIFR–Visiting Summer Research students (June 2016)

News coverage and appearances:

- *Super sharp images with EVN reveal another Fast Radio Burst Coming from a Hypernebula.* [JIVE; 29-Nov-2023][ASTRON; 01-Dec-2023][Universe Today; 05-Dec-2023]
- *Astronomers caught a potent radio burst blasting at us from a dwarf galaxy 3 billion light-years away* [**Popular Science**; 09-June-2022]
- *AstroSat Catches Nuclear Reactions Spreading Across a Neutron Star* [**The Wire**; 07-October-2021]
- *Long burst short: an unexpected supernova* [**Research Matters**; 26-July-2021]
- *Fast Radio Bursts from globular clusters and what could power them* [**National Geographic**; 27-May-2021]
- *Digging the grave: In search of the remains of merging black holes and neutron stars* [**Research Matters**; 14-Sept-2021]
- *FRBs from Galactic magnetar* [**The Wire**; 03-June-2020] [AAS Nova; 09-Sept-2020]
- *Black hole bends light on itself* [**Gizmodo**; 10-April-2020] [**Caltech News**; 08-April-2020] [**SYFY**; 09-July-2020]